POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2017
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Turing test? Explain in detail. 7
   b) Explain about the importance and applications of Artificial Intelligence. 8
2. a) Discuss Crypto Arithmetic Problem. Solve the following Constraint Satisfaction Problem. 7
   SEVEN + EIGHT = TWELVE
   b) Informed search is different from uniformed search? Compare and contrast Breadth First Search with Depth First Search. 8
3. a) Explain the limitations of hill climbing and its solution with appropriate examples. 7
   b) Assume the following facts: 8
   Lionel Messi is a footballer.
   Lionel Messi plays for Barcelona.
   Barcelona is an' A 'Division Spanish Club.
   All the 'A' Division Spanish Clubs play La Liga.
   Use Resolution to prove that "Lionel Messi Plays La Liga"
4. a) Differentiate between Semantic Network and Frames. Draw semantic network of following clauses 8
   Subset_of(People, Mammal), Subset_of(Male, People), Subset of(Female, People), Has_Father(People, Male), Member_of(Ram, Male), Member_of(Sita, Female), Wife_of(Sita, Ram)
   b) What is machine learning? Explain in detail the concept of learning by induction with an example. 7
5. a) How does a supervised learning differ from an unsupervised learning? Explain, Nearest Neighbor learning method with example. 8

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b) Explain the reinforcement learning with example.

6. a) How does the perceptron learn its classification tasks? Explain.
   b) Explain the steps involved in Natural Language Processing.

7. Write short notes on: (**Any two**)
   a) Application of Baysian networks for reasoning
   b) MYCIN
   c) Hopfield network

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